Service-Based Architecture

Software Architecture

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March 16, 2026

Definition 0. Distributed System

A system with multiple components located on different machines that communicate and coordinate actions in order to appear as a single coherent system to the end-user.

- Introduce idea of distributed systems
- Service-based is a simple distributed architecture

Quote

A distributed system is one in which the failure of a computer you didn't even know existed can render your own computer unusable.

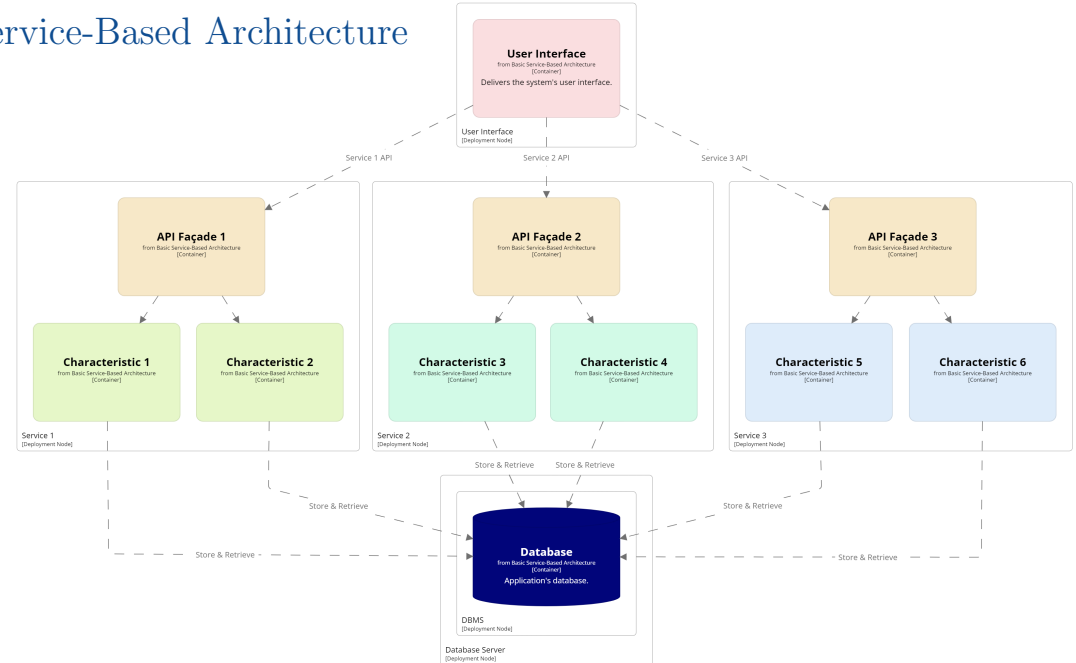
– Leslie Lamport [Turing Award, 2013]

Definition 0. Service-Based Architecture

System is partitioned into business domains that are deployed as distributed services. Functionality is delivered through a user interface that interacts with the domain services.

Explain why this leads to a fairly simple distributed architecture

Service-Based Architecture



Terminology

User Interface Provides access to system functionality

Services Implement functionality for a single, independent business process

Service APIs Communication mechanism between UI and each service

Database Stores persistent data for the system

- Service APIs are communication protocols and data formats, not just a Java-style interface
- Usually all Service APIs use the same communication protocol (e.g. REST)
- Messages between the UI and services are typically asynchronous

Definition 0. API Abstraction Principle

Services should provide an API that hides implementation details.

- Each service publishes its own API
- Hides service implementation details, reducing coupling between UI and service
- Makes it easier to reuse service across systems or by supporting service (e.g. auditing)

Definition 0. Façade Design Pattern

Provide a simple, abstract interface to use a service domain's functionality. A component within the service coordinates how to deliver the requested functionality with the service's internal components.

- Summarise *Façade Design Pattern* and how it is used in a service-based architecture
- Mention its from the GoF book

Definition 0. Independent Service Principle

Services should be independent, with no dependencies on other services.

Explain consequences of dependencies between services

- Services can't easily be deployed separately if they depend on other services
- Would require interfaces between services, increasing coupling

Question

What are the consequences of having a shared database?

Question

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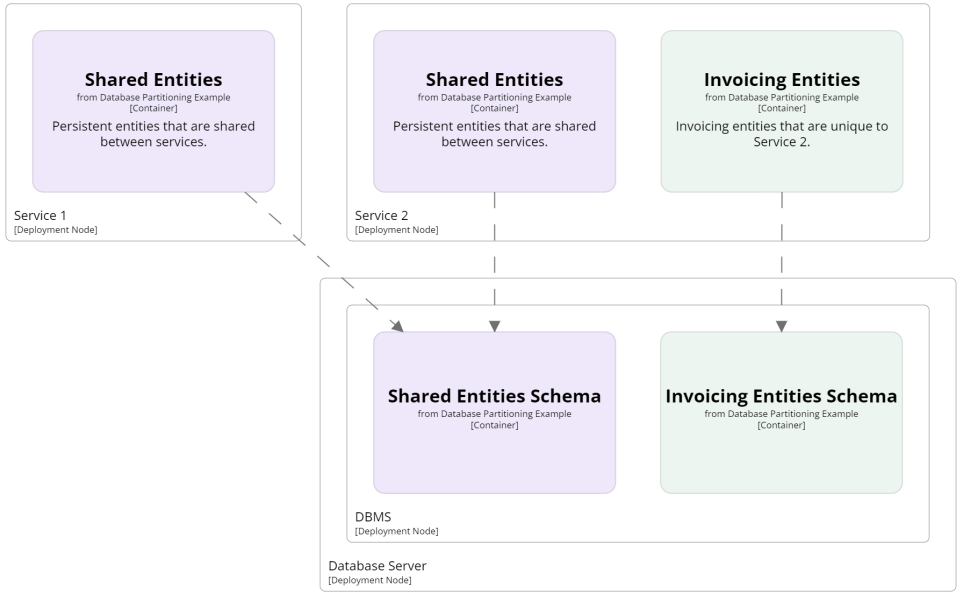
Answer

Increased *data coupling*

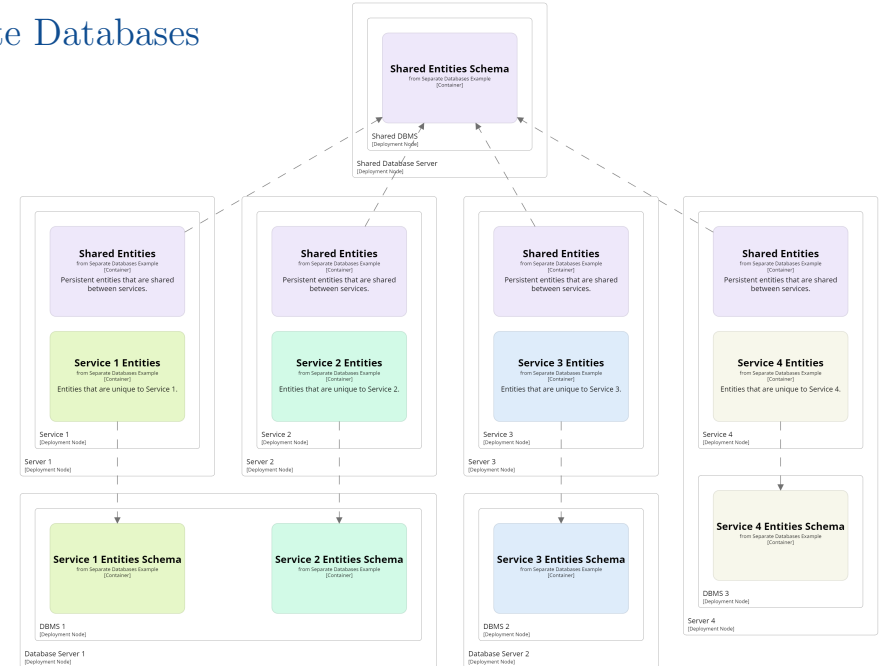
- If a row of a database is locked and another service wants to use it, it is blocked. Losing efficiency benefits of a distributed system.
- If one service changes the structure of its persistent data, all services using that data need to be updated and tested.
- If one service changes how it uses persistent data, all other services using the same data need to be retested.

Logical Partitioning of Persistent Data



- Define a minimal set of shared persistent objects
- Create a shared library to access these objects
- Changes to shared persistent objects are restricted as they require changes to other services
- Each service may have its own persistent objects stored in tables that are not shared with other services

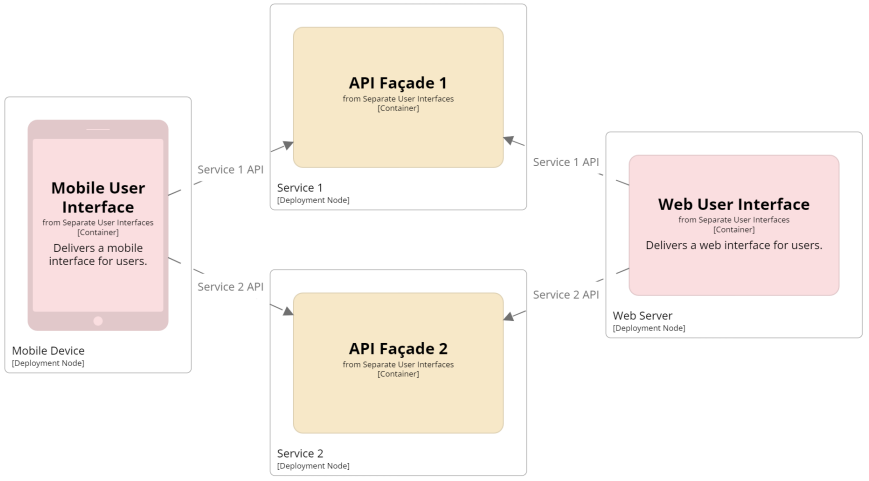
Separate Databases



Discuss options of

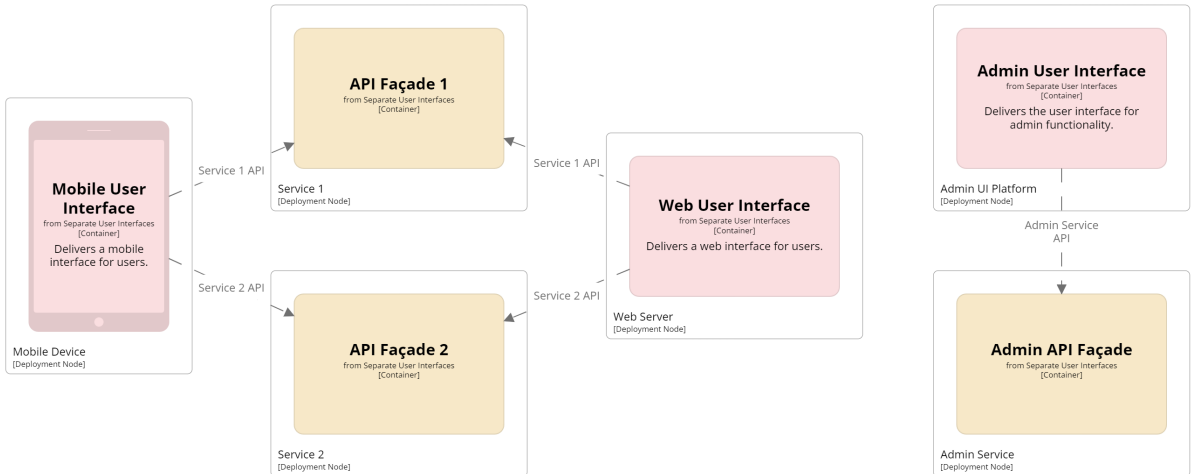
- separate DB servers,
- multiple DBs on one server,
- DBs embedded in application.

Separate UIs



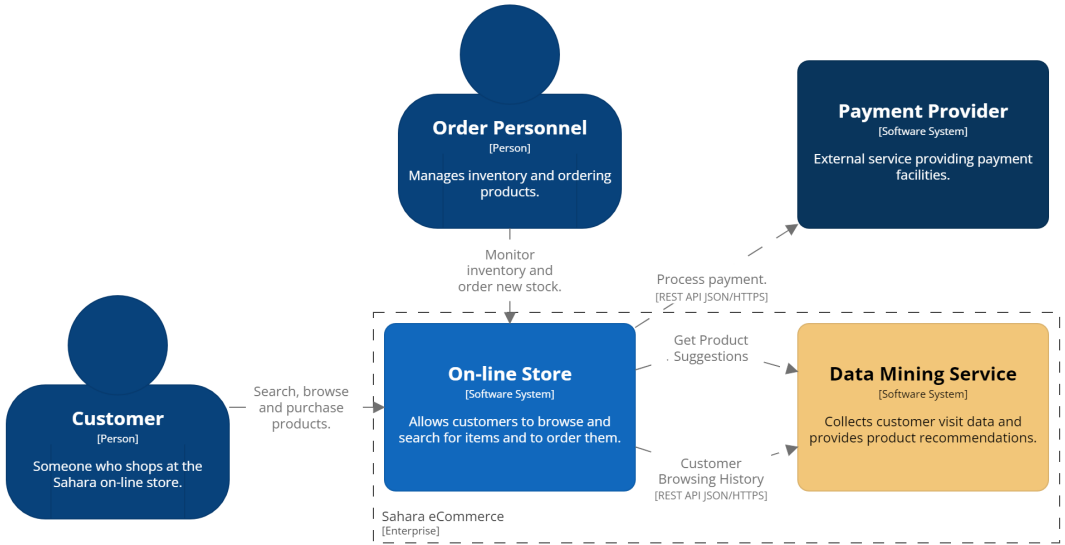
- UI Platform could be desktop, web or mobile app
- Allows multiple concurrent users, even through one user interface

Separate UIs



- UI Platform could be desktop, web or mobile app
- Allows multiple concurrent users, even through one user interface
- Allows *separate UIs* for different clients or roles

Sahara: Context Diagram



- Summarise Sahara eCommerce example
- Order Personnel & Payment Provider added to this example

On-line Store Service Domains

Browsing Customers can find products & add to cart

Purchasing Customers can purchase products in cart

Fulfilment Customers & staff can track order fulfilment

Account Management Customers can manage their account details

Inventory Management Staff can view stock levels and order new stock

- Service-based architectures are based on domain partitioning
- Fulfilment domain behaviours
  - Customer's track order
  - Staff generate pick lists and packaging details
- Inventory management domain behaviours
  - Generate reports on stock levels and product popularity
  - Ordering new stock

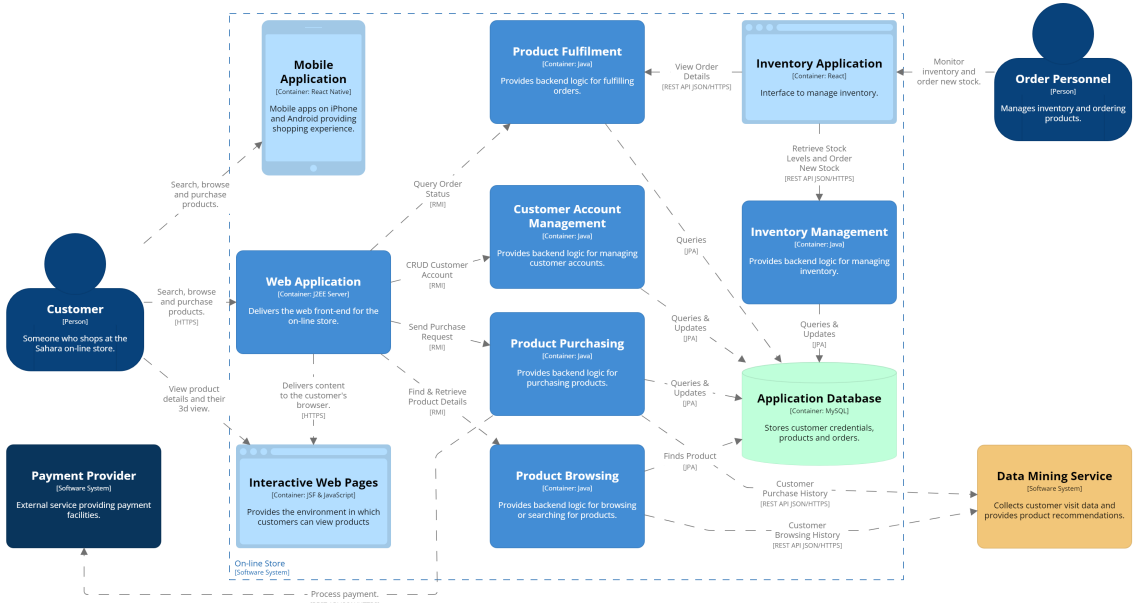
Partitioning

Services are defined by domain partitioning

Coarse Services

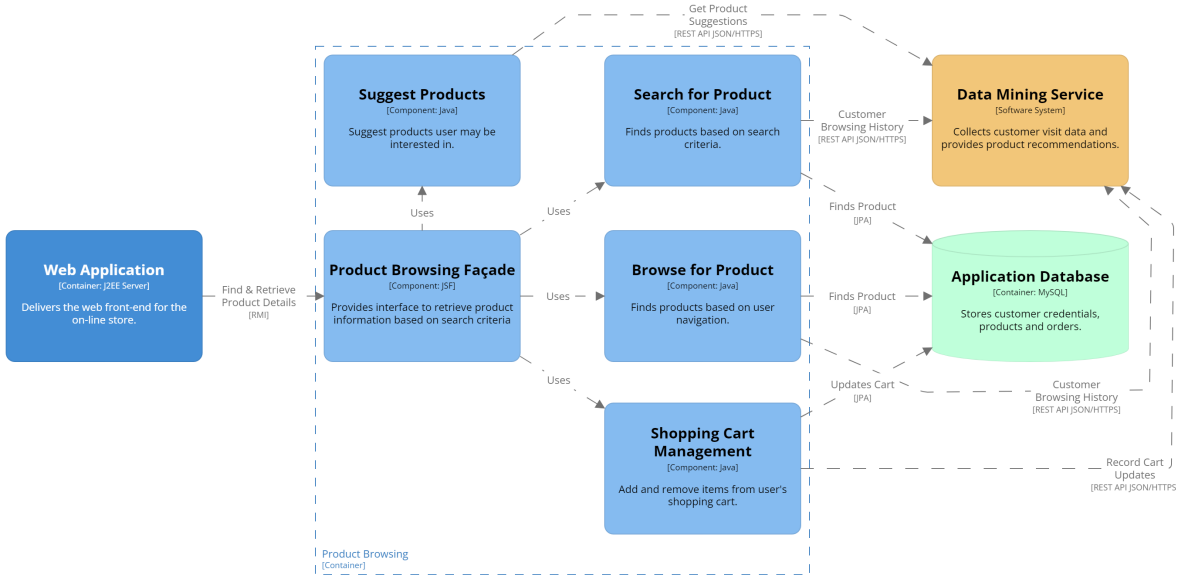
- Domains are large
  - *Coarse-grained* services
- Each service has an internal architecture
  - Technical or domain partitioning

Sahara: On-line Store Container Diagram



- Review Domain Services in context of the overall system
- Mobile App relationships not shown to reduce clutter
- Single DB for simplicity of diagram, not good practice, should be split
- Cart shared with Browsing & Purchasing
- Order shared with Purchasing & Fulfilment
- Product shared with everything, except Account Mgt
- User shared with almost everything
- Most of these will be query only, or only locking a single row
- Repeat idea that Service APIs mean you may have multiple UIs (Web, Mobile, Inventory Apps)
- *Inventory App* is an example of a separate UI delivering a different app

Sahara: Product Browsing Component Diagram



- Summarise the components making up the key parts of the Product Browsing Service (container)
- Product Browsing Façade provides the Service API

Product Browsing Service API

Search https://api.sahara.com/v1/search?keywords=...

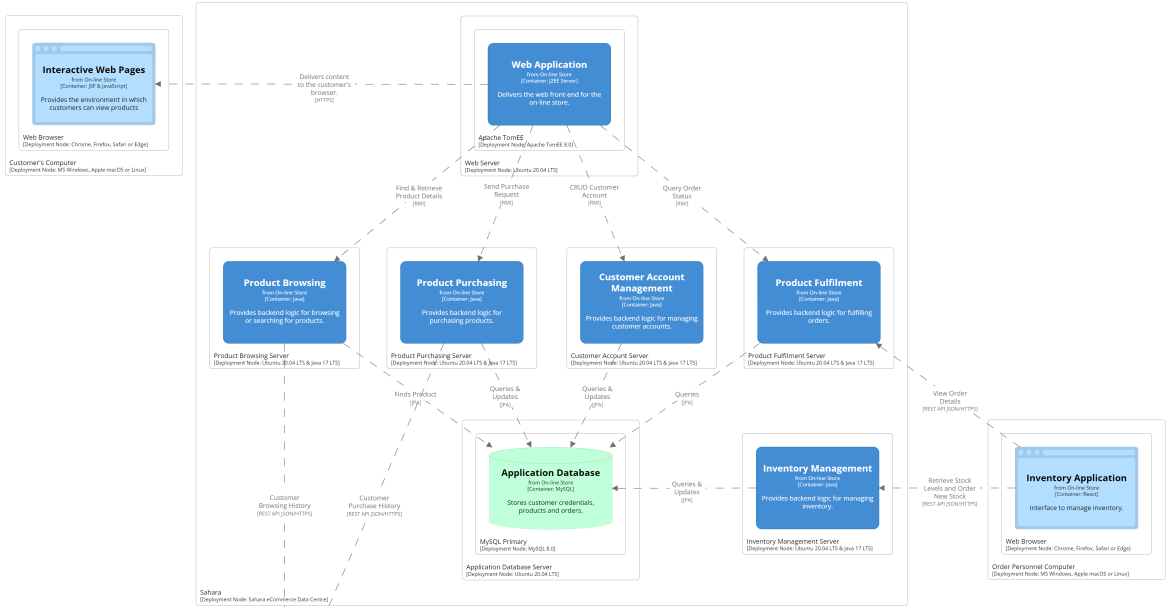
Browse https://api.sahara.com/v1/browse?category=...

Add to Cart https://api.sahara.com/v1/cart

- JSON to pass data
- JSF action controller handles request

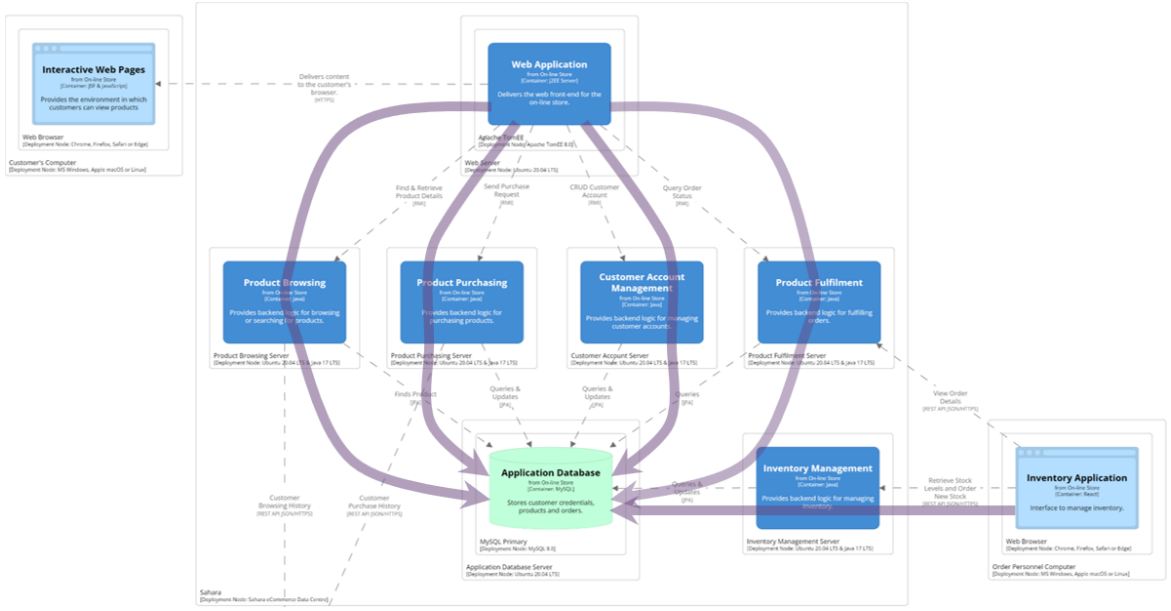
- Search & Browse are GET requests passing parameters
- Add to Cart is a POST request passing product id and quantity to be added to cart
- Authentication needs to be part of requests
- API Versioning shown in URIs

Sahara: Deployment Diagram



- UI Platform could be desktop, web, mobile app, VR, ...
- Allows multiple concurrent users, even through one UI

Sahara: Concurrent Access



- Emphasise many users from different UIs accessing distributed services concurrently
- Point out that this & REST require stateless services

Question

If the probability of something happening is one in 10^{25}, how often would it happen?

- Common sense says, "essentially never"
- Physicist might say, "*All the time* – in a cubic metre of air, molecules are colliding all the time"

Assume there will be failures in a distributed system

Design to manage them

Question

What happens if a service goes down?

Question

What happens if a service goes down?

Answer

Need to manage timeouts, retries, graceful failure, ...

- Some of this can be managed by infrastructure
  - Requires monitoring systems
- Some issues are harder to deal with due to coarse service domains
- Some of this needs to be managed within the application

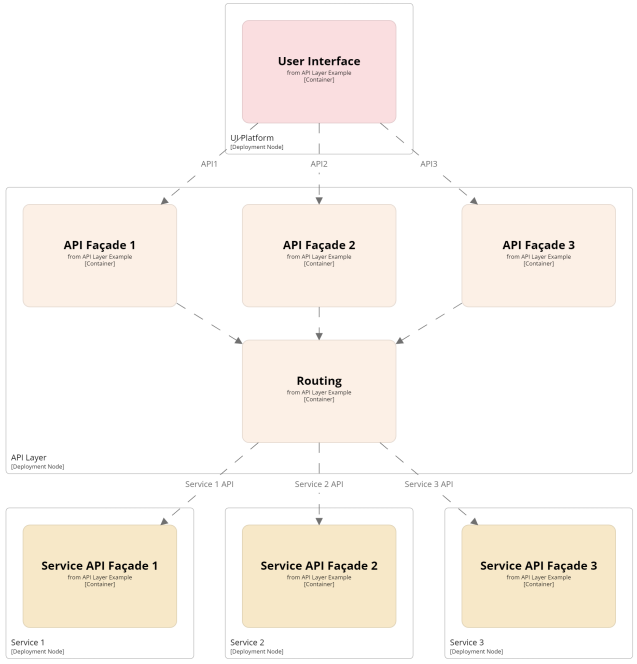
Consider Network Failure

If customer tried to add product to cart:

- What happens if Product Browsing didn't receive it?
- What happens if UI didn't get a response?
- What happens if Database wasn't updated?

- Network failure
  - before Product Browsing receives msg
  - after Product Browsing saves state, but before UI receives response
  - when communicating with DB
- DB update fails but doesn't notify Product Browsing
- The more interactions between systems, the more things you have to think about recovering from

API Layer


User Interface
from API Layer Example
[Container]
UI-Platform
[Deployment Node]
API1
API2
API3
API Façade 1
from API Layer Example
[Container]
API Façade 2
from API Layer Example
[Container]
API Façade 3
from API Layer Example
[Container]
Routing
from API Layer Example
[Container]
API Layer
[Deployment Node]
Service 1 API
Service 2 API
Service 3 API
Service API Façade 1
from API Layer Example
[Container]
Service API Façade 2
from API Layer Example
[Container]
Service API Façade 3
from API Layer Example
[Container]
Service 1
[Deployment Node]
Service 2
[Deployment Node]
Service 3
[Deployment Node]

API Layer Advantages

- Acts as a reverse proxy or gateway to services
- Hides internal network structure
- Easier to implement *cross-cutting* concerns
  - e.g. security policies
- Allows service discovery
  - Interface to register service
  - Clients can find out what services are available

- *Reverse proxy* hides internal network structure of architecture
  - Can expose different interfaces to external & internal systems
  - Facilitates delivering the security principle of least privilege
- *Gateway* adds "intelligence" to the reverse proxy
  - Can process requests & responses
  - Orchestrate or aggregate requests / responses to improve performance
  - Translate protocols

Pros & Cons

- Simplicity *For a distributed system* 
- Modularity Services 
- Extensibility New services 
- Deployability Independent services 
- Testability Independent services 
- Security API layer 
- Reliability Independent services 
- Interoperability Service APIs 
- Scalability Coarse-grained services